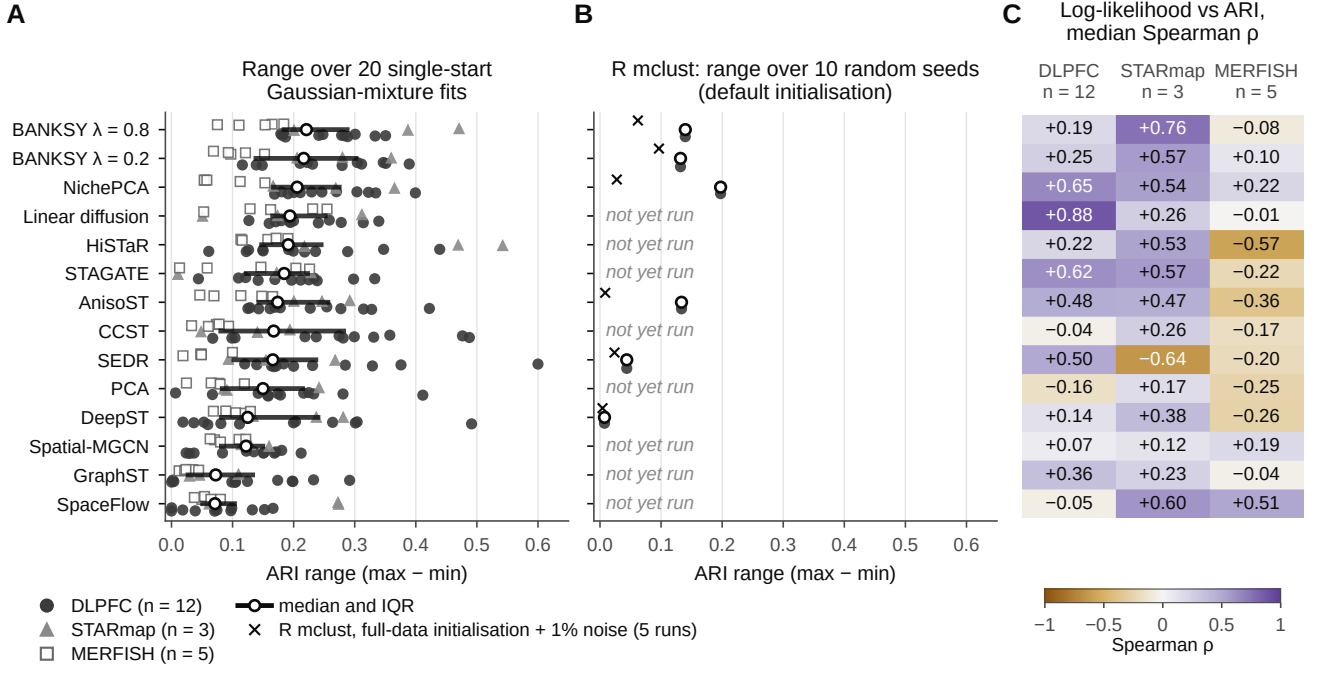

A
Range over 20 single-start Gaussian-mixture fits
BANKSY λ = 0.8
BANKSY λ = 0.2
NichePCA
Linear diffusion
HiSTaR
STAGATE
AnisoST
CCST
SEDR
PCA
DeepST
Spatial-MGCN
GraphST
SpaceFlow
0.0 0.1 0.2 0.3 0.4 0.5 0.6
ARI range (max − min)
B
R mclust: range over 10 random seeds (default initialisation)
not yet run
0.0 0.1 0.2 0.3 0.4 0.5 0.6
ARI range (max − min)
C
Log-likelihood vs ARI, median Spearman ρ
DLPFC n = 12
STARmap n = 3
MERFISH n = 5
+0.19 +0.76 −0.08
+0.25 +0.57 +0.10
+0.65 +0.54 +0.22
+0.88 +0.26 −0.01
+0.22 +0.53 −0.57
+0.62 +0.57 −0.22
+0.48 +0.47 −0.36
−0.04 +0.26 −0.17
+0.50 −0.64 −0.20
−0.16 +0.17 −0.25
+0.14 +0.38 −0.26
+0.07 +0.12 +0.19
+0.36 +0.23 −0.04
−0.05 +0.60 +0.51
−1 −0.5 0 0.5 1
Spearman ρ
DLPFC (n = 12)
STARmap (n = 3)
MERFISH (n = 5)
median and IQR
R mclust, full-data initialisation + 1% noise (5 runs)